higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTIONS
NQF LEVEL 2

12 MARCH 2019

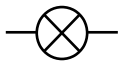
This marking guideline consists of 7 pages.

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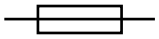
QUESTION 1

1.1 •Underground
•Aerial (overhead) (2)

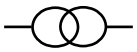
1.2 1.2.1 Lamp/bulb

1.2.2 

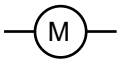
1.2.3 Fuse

1.2.4 

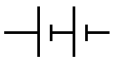
1.2.5 Transformer

1.2.6  or 

1.2.7 Motor

1.2.8 

1.2.9 Battery

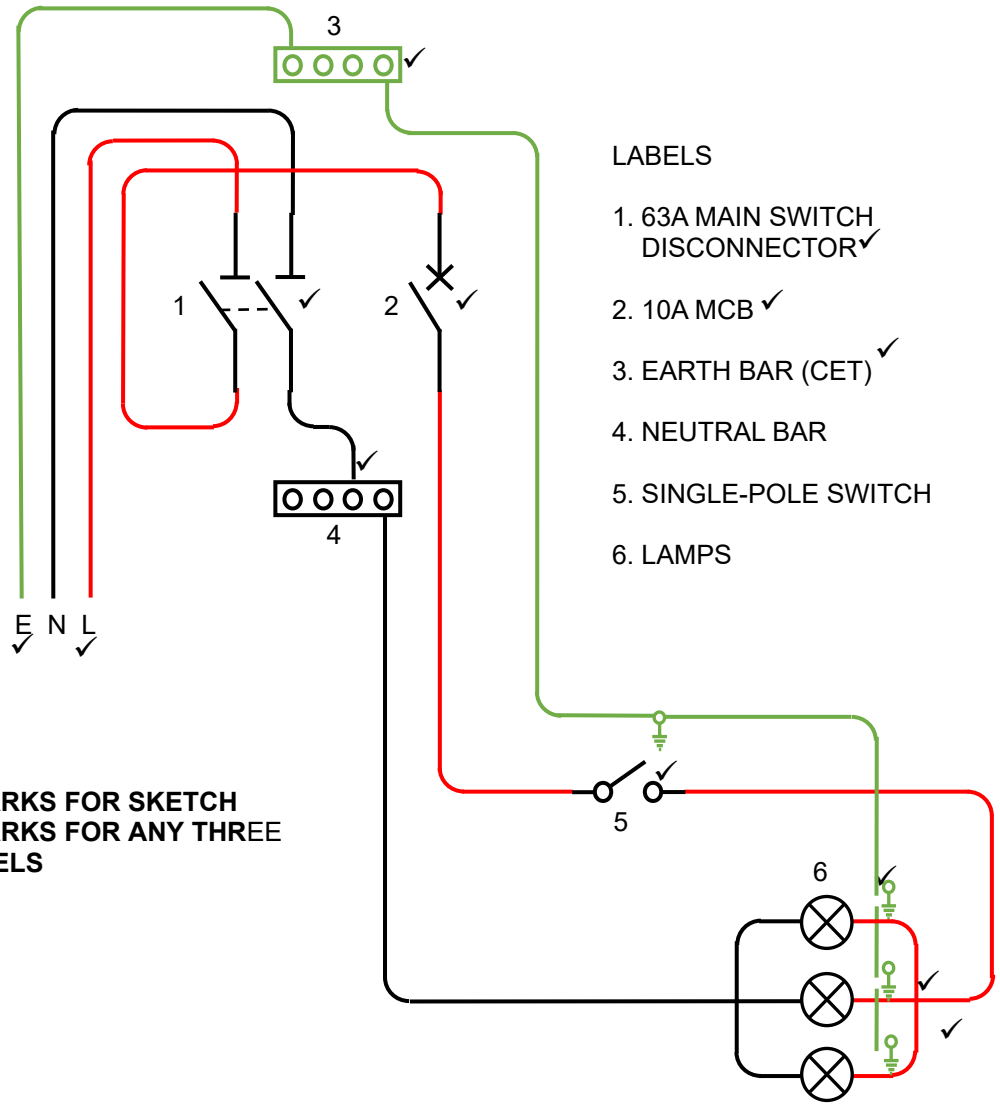
1.2.10 

(10 × 1) (10)

1.3 No.✓ According to SANS 10142-1 the minimum insulation resistance value recorded may not be less than 1 MΩ.✓ (2)

1.4 Insulation resistance tester/megger (1)

1.5



7 MARKS FOR SKETCH
3 MARKS FOR ANY THREE LABELS

FIGURE 1

(10 × 1) (10)
[25]

QUESTION 2

2.1 2.1.1 True
2.1.2 True
2.1.3 False
2.1.4 False
2.1.5 True
(5 × 1) (5)

2.2 2.2.1 Ohmmeter/Multimeter (1)

2.2.2 Lowest ohm scale (1)

2.2.3 Place test leads✓ on each end of the conductor.✓ (2)

2.2.4 No.✓ It indicates an open circuit on the conductor.✓ (2)

2.3
- Trunking is neat
- Conductors are laid in
- Easy access to wiring
- Easy to add or remove conductors
- Not necessary to pull conductor or cabling through (Any 4 × 1) (4)

2.4
- Strip the cable ends to the required length.
- Join the conductors using ferrules.
- Also join the steel wire armouring.
- Place the mould over the joint and seal the ends.
- Mix and pour the resin into the mould. (5)

[20]

QUESTION 3

3.1 A – Frame/Yoke
B – Main pole
C – Field winding
D – Pole shoe
E – Brush
F – Armature core
G – Commutator
H – Armature windings
(8 × 1) (8)

3.2 It is used to reduce sparking✓ at the brushes and commutator.✓ (2)

3.3
- Check that the shaft turns freely by hand.
- Check the shaft bearing and end play by moving it sideways.
- Check that the keyway on the shaft is not damaged. (3)

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3.4 •Foundation studs
•Slide rails
•Metallic bases (Any 2 × 1) (2)
[15]

QUESTION 4

4.1 4.1.1
- Press to test
- Sensitivity test (2)

4.1.2 Earth leakage relay and circuit breaker tester (1)

4.1.3 No
The earth leakage relay should trip at not more than 30 mA. (2)

4.2 4.2.1 Ohmmeter/Multimeter on lowest ohm range
4.2.2 Insulation resistance tester/megger
4.2.3 Hand
4.2.4 Ammeter/Tong tester
4.2.5 Ohmmeter/Multimeter on lowest ohm range
(5 × 1) (5)

4.3
- Switch off the power to the geyser circuit (isolate).
- Apply lockout and tagout procedures.
- Test to ensure the power is off.
- Set the multimeter to the lowest ohm scale.
- Place the test leads on the terminals of the element and record the value. (5)

[15]

QUESTION 5

5.1 Timer (1)

5.2 •Analogue
•Digital (2)

5.3

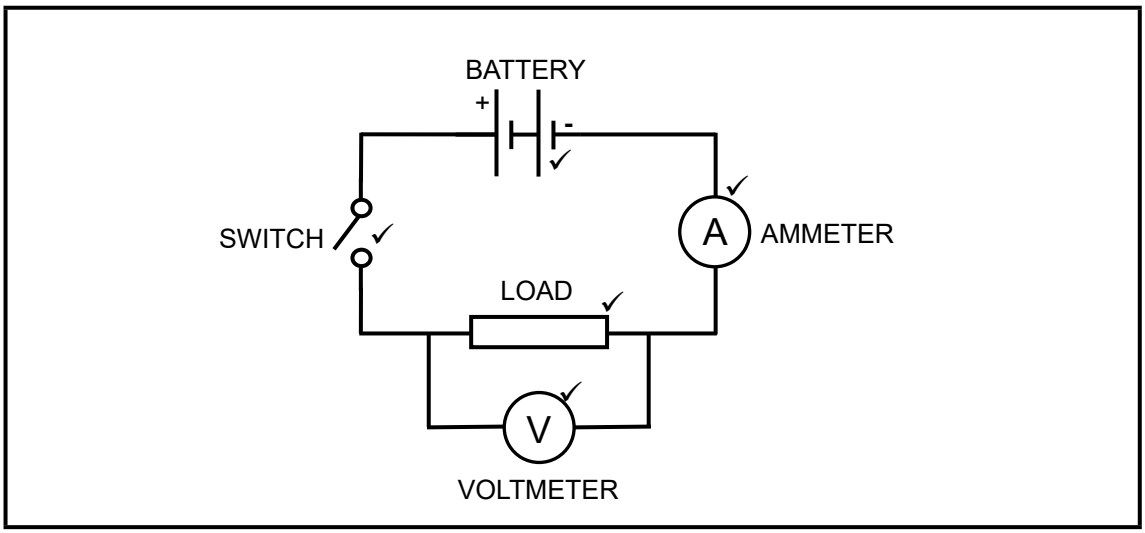


FIGURE 2
5 MARKS FOR CORRECT CONNECTION OF COMPONENTS
2 MARKS FOR ANY TWO CORRECT LABELS (7)

5.4
- Voltmeter/Multimeter
- Screwdrivers
- Side cutter
- Combination pliers
- Crimping pliers
- Cable knife (Any 5 × 1) (5)

[15]

QUESTION 6

6.1	6.1.1	C		
	6.1.2	E		
	6.1.3	D		
	6.1.4	G		
	6.1.5	A		
			(5 × 1)	(5)
6.2	6.2.1	Parallel		(1)
	6.2.2	$I_T = I_1 + I_2$ $I_T = 0,5 + 0,5$✓ $I_T = 1$ A✓		(2)
	6.2.3	$V_T = V_1 = V_2$✓ $V_T = 12$ V✓		(2)
				[10]
			TOTAL:	**100**

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